



Engineering Graphics

Engineering graphics is the language used by engineers to communicate shape, size, construction and manufacturing information. This handbook develops standards-based drawing habits, spatial visualisation and the ability to convert three-dimensional objects into accurate two-dimensional views.

REV2026

Semester I

Drawing

Malayalam support

Printable handbook

Course code

1003

Semester

I

Course type

Drawing

Contact hours

45

Credits

1.5

CIE / ESE

40 / 60

Modules

4

COURSE OVERVIEW

What this handbook covers

The structure follows the supplied official SITTTTR Kerala Revision 2026 syllabus. It is a learning handbook rather than a replacement for the official syllabus PDF. Use the module checklist to ensure that every prescribed point has been revised.

CO1

Illustrate the basic elements, standards and conventions of engineering drawing.

CO2

CO3

Interpret and draw projections of points and lines.

CO4

Develop orthographic projections and full sectional views of simple engineering objects.

Module roadmap

Module I

Introduction — terminology, dimensioning and polygons

Module II

Basic figures — conics, involutes and developments

Module III

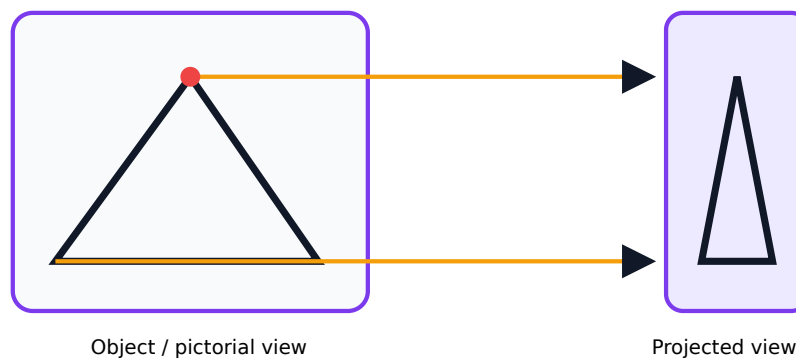
Basic projections — points and lines

Module IV

3D projections — orthographic and sectional views

VISUAL CONCEPT

See the subject as a system



MODULE I

Introduction — terminology, dimensioning and polygons

Engineering Drawing-ൽ line type, lettering, dimensioning, symbol എന്നിവ ഒരു standard പ്രകാരം ഉപയോഗിക്കണം. Drawing മനോഹരമാക്കുന്നതിനേക്കാൾ പ്രധാനമാണ് അത് ambiguity ഇല്ലാതെ manufacture ചെയ്യാൻ കഴിയുന്ന രീതിയിൽ വായിക്കാനാകുന്നത്.

Drawing terminology and BIS conventions

Drawing sheet sizes, title block, scales, lettering, visible line, hidden line, centre line, dimension line and extension line. Keep line weights consistent and never use object lines as dimension lines.

Principles of dimensioning

Use one unit system, place dimensions outside the view where possible, avoid duplicate dimensions and dimension to functional features. Explain aligned and unidirectional systems and the correct use of diameter, radius, square and angle symbols.

Scales — study purpose

Recognise representative fraction, full-size, reducing and enlarging scales. Plain, diagonal and vernier scales are included for conceptual study; the drawing should clearly state the scale used.

Regular polygons

Construct pentagon, hexagon, heptagon and octagon when side length is given. Use the general polygon method or a suitable exact construction and verify equal sides before darkening.

Rules / common mistakes

- Dimensions must be sufficient but not repeated.
- Arrowheads must touch extension lines and remain uniform.
- Construction lines are thin; final object outlines are thick.
- Never measure from a printed drawing when a numerical dimension is given.

Practice and self-learning

- Prepare an A3/A4 sheet with title block and standard line samples.
- Redraw a dimensioned component using BIS conventions.
- Construct four regular polygons from a given side.

MODULE II

Basic figures — conics, involutes and developments

Conic Section എന്നത് cone-നെ plane കൊണ്ട് cut ചെയ്യുമ്പോൾ ലഭിക്കുന്ന curve ആണ്. Development of Surface എന്നത് 3D object-ന്റെ surface cut ചെയ്ത് flat sheet ആയി തുറക്കുന്ന രീതിയാണ്; sheet-metal work-ൽ ഇത് വളരെ പ്രധാനമാണ്.

Ellipse — rectangular method

Draw the bounding rectangle using major and minor axes, divide the semi-axes equally, project the divisions and obtain points of the ellipse. Join points using a smooth curve, maintaining symmetry about both axes.

Ellipse — concentric-circle method

Draw concentric circles with radii equal to the semi-major and semi-minor axes. Divide both circles into equal radial parts; combine vertical and horizontal projectors to locate ellipse points.

Parabola — tangent method

Construct the axis, base and vertex; divide the base and axis appropriately, draw tangent-related construction lines and obtain the parabolic curve. State focus/directrix terminology even when the tangent method is used.

Involute of circle and rectangle

An involute is the locus of the end of a taut string unwound from a curve. Divide the perimeter, draw successive tangents and set off cumulative lengths to locate the involute.

Development of prism and cylinder

Use the parallel-line method. Stretch out the base perimeter, draw generators and transfer heights. Include seam allowance only when asked; preserve true lengths.

Development of a 90° elbow

Develop a two-piece symmetric elbow with equal circular sections and standard side outlets. Divide the half-circle, project generators to the miter line and transfer ordinates to the stretch-out.

Rules / common mistakes

- Use a French curve or smooth freehand curve; do not join conic points with straight segments.
- Generators must be numbered consistently in elevation and development.
- A development requires true lengths, not apparent lengths.

Practice and self-learning

- Construct one ellipse by both prescribed methods and compare point accuracy.
- Draw involutes of a circle and a rectangle.
- Prepare developments of a closed rectangular prism, cylinder and two-piece elbow.

Basic projections — points and lines

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Projection എന്നത് object-ൽ നിന്ന് imaginary projectors reference plane-ലേക്ക് വീടുന്ന പ്രക്രിയയാണ്. HP, VP എന്നിവ fold ചെയ്ത ശേഷം XY line-ന്റെ relative position നോക്കിയാണ് point ഏത് quadrant-ലാണെന്ന് കണ്ടെത്തുന്നത്.

Theory of projection

Define projector, plane of projection, horizontal plane (HP), vertical plane (VP), profile plane and XY reference line. Compare first-angle and third-angle projection and identify the standard symbol.

Projection of points in four quadrants

Locate front view from distance above/below HP and top view from distance in front of/behind VP. Use sign discipline and place views on the correct side of XY.

Lines parallel to both planes

A line parallel to HP and VP shows true length in both front and top views. End projectors remain aligned and distances from XY stay constant.

Lines perpendicular to one plane

A line perpendicular to a plane appears as a point in that plane and as true length in the other view.

Lines inclined to one plane

For first-quadrant problems, draw the view parallel to the plane first at true length and given angle, then project the shortened view on the other plane.

Rules / common mistakes

- Projectors are thin and perpendicular to XY.
- Label corresponding ends with the same lowercase letter and prime convention.
- Read wording carefully: inclination to HP is seen in elevation only when the line is parallel to VP.

Practice and self-learning

- Prepare a quadrant chart showing the position of front and top views.
- Solve one point problem in each quadrant.
- Draw line cases: parallel to both, perpendicular to one, inclined to HP, inclined to VP.

3D projections — orthographic and sectional views

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CO4

Orthographic Projection-ൽ 3D object-നെ mutually perpendicular directions-ൽ നോക്കി Front, Top, Side views വരയ്ക്കുന്നു. Sectional View-ൽ imaginary cutting plane ഉപയോഗിച്ച് hidden internal details വ്യക്തമായി കാണിക്കുന്നു.

Pictorial and orthographic views

Distinguish isometric view from orthographic views. Select the front view that shows maximum shape information and minimum hidden detail.

First-angle arrangement

In first-angle projection, the top view is placed below the front view and the right-side view is placed to the left. Use a 45° miter line when transferring depth to side view.

View correlation

All views must align: widths between front and top, heights between front and side, and depths between top and side. Project before darkening.

Hidden and centre details

Use dashed hidden lines only when they add necessary information. Draw centre lines for axes, holes and symmetry; extend centre lines slightly beyond the feature.

Full sectional view

Pass a cutting plane through the object, remove the portion in front and hatch the cut material at a uniform angle and spacing. Do not hatch holes, ribs cut longitudinally, shafts or standard fasteners unless specified.

Sectional orthographic problems

Develop front, top and side views of simple engineering objects and produce full sections from pictorial views using first-angle projection.

Rules / common mistakes

- Choose the minimum number of views needed to define the object.
- Hatching of adjacent parts must differ in angle or spacing.
- Do not show hidden lines in a sectional view unless essential.

Practice and self-learning

- Convert five simple isometric objects into three orthographic views.
- Draw a full sectional front view and corresponding top view.

- Perform a final drawing-sheet audit for alignment, line type, labels and dimensions.

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QUICK REVISION

Formula / rule bank

Rule / Formula 1

BIS/ISO line conventions and first-angle symbol

Rule / Formula 2

Aligned and unidirectional dimensioning

Rule / Formula 3

Ellipse: rectangular and concentric-circle methods

Rule / Formula 4

Parabola: tangent method

Rule / Formula 5

Involute of circle/rectangle

Rule / Formula 6

Parallel-line development

Rule / Formula 7

Projection of points in all quadrants

Rule / Formula 8

Line inclined to one plane only

Rule / Formula 9

First-angle orthographic arrangement

Rule / Formula 10

Full sectional view and hatching rules

Worked checkpoints

Worked checkpoint 1

Construct a regular hexagon of side 35 mm and verify all diagonals before darkening.

Worked checkpoint 2

A point is 25 mm above HP and 30 mm in front of VP: place elevation 25 mm above XY and plan 30 mm below XY.

Worked checkpoint 3

A 60 mm line parallel to VP and inclined 30° to HP: draw elevation at 30° with true length, then project the plan.

Worked checkpoint 4

Develop a closed cylinder of diameter 50 mm and height 80 mm using 12 equal divisions.

APPLICATION

Practical, drawing and self-learning work

No	Task	Required evidence
1	Sheet 1: line types, lettering and dimensioning standards.	Record procedure, observation/result and safety/quality checks.
2	Sheet 2: pentagon, hexagon, heptagon and octagon.	Record procedure, observation/result and safety/quality checks.
3	Sheet 3: ellipse by rectangular and concentric-circle methods; parabola by tangent method.	Record procedure, observation/result and safety/quality checks.
4	Sheet 4: involute of circle and rectangle.	Record procedure, observation/result and safety/quality checks.
5	Sheet 5: development of prism, cylinder and 90° elbow.	Record procedure, observation/result and safety/quality checks.
6	Sheet 6: points in all four quadrants.	Record procedure, observation/result and safety/quality checks.
7	Sheet 7: projection of lines in first quadrant.	Record procedure, observation/result and safety/quality checks.
8	Sheet 8-10: orthographic and full sectional views.	Record procedure, observation/result and safety/quality checks.

EXAM PREPARATION

Expected questions and model-answer guides

These are syllabus-aligned practice questions, not a claim about the exact end-semester paper.

2 marks

Q1. List the standard line types and state one use of each.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q2. State four rules of dimensioning.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q3. Construct a regular heptagon when side length is given.

Model-answer guide

Course 1003

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q4. Explain the rectangular method of drawing an ellipse.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q5. Explain how an involute of a circle is constructed.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q6. Describe the development of a two-piece 90° elbow.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q7. Compare first-angle and third-angle projection.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q8. Draw projections of a point in each quadrant.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q9. Explain the method for a line inclined to HP and parallel to VP.

Model-answer guide

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Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q10. Convert a given isometric object into front, top and side views.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q11. State the rules for hatching in sectional views.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q12. Draw a full sectional front view of a simple object.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

COVERAGE AUDIT

Official syllabus coverage checklist

- Module I: Drawing terminology and BIS conventions
- Module I: Principles of dimensioning
- Module I: Scales — study purpose
- Module I: Regular polygons
- Module II: Ellipse — rectangular method
- Module II: Ellipse — concentric-circle method
- Module II: Parabola — tangent method
- Module II: Involute of circle and rectangle
- Module II: Development of prism and cylinder
- Module II: Development of a 90° elbow
- Module III: Theory of projection
- Module III: Projection of points in four quadrants
- Module III: Lines parallel to both planes
- Module III: Lines perpendicular to one plane

- Module III: Lines inclined to one plane

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- Module IV: Pictorial and orthographic views
- Module IV: First-angle arrangement
- Module IV: View correlation
- Module IV: Hidden and centre details
- Module IV: Full sectional view
- Module IV: Sectional orthographic problems

REFERENCES

Learning resources

1. Engineering Graphics — P. I. Varghese, VIP Publishers
2. Engineering Graphics — K. C. John, PHI Learning

Primary authority for scope: supplied official SITTTR Kerala Diploma Curriculum Revision 2026 syllabus PDF for Course 1003.